

Course Title: **System Analysis and Design**

Course No. : ICT Ed. 447

Level: Bachelor

Semester: Fourth

Nature of Course: Theoretical + Practical

Credit Hours: 3 (2T+1P)

Teaching Hours: 64 (32T+32P)

1. Course Description

The course is a blend of understanding of system analysis & design with its practical applications. This course includes understanding of various elements of system analysis and design with emphasis on the application of information technology issues as a business tool. The course covers components of system analysis and design techniques, data modeling, logical process modeling, and object oriented modeling techniques.

2. Course Objectives

Following are the general objective of this course:

- To familiarize the students with System Development Life Cycle.
- To enable student to recognize different system development models
- To enhance the skill of students in System analysis and design of user centered System requirements.
- To make the students competent in analysis, design and implementation.
- To make the students knowledgeable about the latest trends of modern system analysis and design.

3. Specific Objectives and Contents

Specific Objectives	Contents
• Explain system and its major components • Identify system attributes, properties and characteristics • Define System Development Lifecycle • Discuss steps in System Development Lifecycle • Understand the user needs • Differentiate between commercial product and contract system development	Unit 1: System: attributes, properties and characteristics (3 Hrs) 1.1. What is a System? 1.2. Learning to Recognize Types of Systems 1.3. Analytical Representation of a System 1.4. System Stakeholders: User and End User Roles 1.5. System Attributes, Properties and Characteristics 1.6. System/Product Life Cycle Concepts 1.7. Understanding the User's Problem, Opportunity, and Solution Spaces 1.8. Commercial/Consumer Product Versus Contract System Development <u>Practical Works</u> (3 hrs) • Visit any type of organization and observe their business process and prepare a field report that identifies each and every aspect of their business process and present it in your class too.

• Explain and understand different system development models: waterfall, spiral, iterative and incremental, evolutionary, agile development model, • Differentiate between different system development models	Unit 2: System Development Process Model (5 Hrs) 2.1. Introduction to the System Development Models 2.2. Waterfall Development Strategy and Model 2.3. “V” System Development Strategy and Model 2.4. Spiral Development Strategy and Model 2.5. Iterative and Incremental Development Model 2.6. Evolutionary Development Strategy and Model 2.7. Agile Development Strategy and Model 2.8. Selection of System Versus Component Development Models <u>Practical Works</u> (5 hrs) • Visit Software Development Company and study or enquire the system development process model used by them and prepare report about it and also present in class.
• Describe importance of system documentation • Know about data accession list (DAL) and data criteria list (DCL) • Create system documentation • Identify different issues in system documentation	Unit 3: System Documentation Strategy (4 Hrs) 3.1. Quality System and Engineering Data Records 3.2. System Design and Development Data 3.3. Data Accession List (DAL) and Data Criteria List (DCL) 3.4. SE and Development Documentation Sequencing 3.5. Documentation Levels of Formality 3.6. Export Control of Sensitive Data and Technology 3.7. System Documentation Issues <u>Practical Works</u> (4 hrs) • Study system document written for any type of system (which can be collected from local or global software company), then share your understanding in class.
• Discuss importance of requirements • Know to gather requirements • Design requirement traceability path • Construct requirement statements • Develop appropriate requirement verification methods	Unit 4: Requirement Derivation, Allocation, Flow Down and Traceability (4 Hrs) 4.1 Introduction 4.2 Requirements Derivation Methods 4.3 Requirements Derivation and Allocation Across Entity Boundaries 4.4 Requirements Allocation 4.5 Requirements Traceability 4.6 Preparing the Requirement Statement 4.7 Selection of Requirement Verification Methods <u>Practical Works</u> (4 hrs)

	Prepare requirement document for the system that you are going to develop in this semester. The system might be the one which was identified in practical works of unit 1.
- Discuss user centered system design - Design system incorporating Human Factors and Ergonomics - Decompose complex system into small chunks to better understand - Apply SE, HF and Ergonomics actions	Unit 5: User Centered System Design (4 Hrs) 5.1 Introduction to UCSD 5.2 Understanding Human Factors (HF) and Ergonomics 5.3 Situational Assessment: Areas of Concern 5.4 Complex System Development 5.5 SE, HF and Ergonomics Actions <u>Practical Works</u> (4 hrs) Read at least one research article related to user centered system design and write review and share with class.
- Demonstrate different system architecture models - Understand and make use of system architecture to design interface - Design interface embedding control challenges	Unit 6: System Architecture Development (4 hrs) 6.1 Introduction to System Architecture Development 6.2 Development of System Architectures 6.3 Interface Definition Methodology 6.4 Interface Design—Advanced Topics 6.5 Interface Definition and Control Challenges and Solutions <u>Practical Works</u> (4 hrs) Prepare architectural design of the proposed system (proposed in practical works of unit 4).
- Discuss system testing - Design test cases - Prepare test data - Perform test procedures - Understand the test challenges and resolve	UNIT 7: System Integration, Test, and Evaluation (SITE) (4 Hrs) 7.1 SITE Fundamentals 7.2 Key Elements of SITE 7.3 Planning for SITE 7.4 Establishing the Test Organization 7.5 Developing Test Cases (TCs) and Acceptance Test Procedures (ATPs) 7.6 Performing SITE Tasks 7.7 Common Integration and Test Challenges and Issues <u>Practical Works</u> (4 hrs) Prepare test case and test data for the system designed in practical works of unit 6.
- Discuss system deployment - Perform system maintenance	Unit 8: System Deployment, OM&S, Retirement, and Disposal (4 Hrs) 8.1 System Deployment Operations

• Understand system reliability, maintainability and availability • Perform system disposal operations	8.2 System Operation, Maintenance, and Sustainment (OM&S) 8.3 System Reliability, Maintainability, and Availability (RMA) 8.4 System Retirement (Phase-Out) Operations 8.5 System Disposal Operations <u>Practical Works</u> (4 hrs) Visit any organization and study the system deployment process used by them and prepare report about it and also present in class.
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9 Instructional Techniques

The instructional techniques for this course are divided into two groups. First group consists of general instructional techniques applicable to most of the units. The second group consists of specific instructional techniques applicable to particular units.

9.1 General Instructional Techniques

Reading materials will be provided to students in each unit. Lecture preferably with the use of multi-media projector, demonstration, practical classes, discussion, and brain storming are used in all units.

4.2 Specific Instructional Techniques

Demonstration is an essential instructional technique for all units in this course during teaching-learning process. Specifically, demonstration with practical works will be specific instructional technique in this course.

10 Evaluation :

Internal Assessment	External Practical Exam/Viva	Semester Examination	Total Marks
40 Points	20 Points	40 Points	100 Points

Note: Students must pass separately in internal assessment, external practical exam and semester examination.

10.1 Internal Evaluation (40 Points):

Internal evaluation will be conducted by subject teacher based on following criteria:

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| 1) Class Attendance | 5 points |
| 2) Learning activities and class performance | 5 points |
| 3) First assignment (written assignment) | 10 points |
| 4) Second assignment (Case Study/project work with presentation) | 10 points |

5) Terminal Examination

10 Points

Total	40 points
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10.2 Semester Examination (40 Points)

Examination Division, Dean office will conduct final examination at the end of semester.

- 1) Objective question (Multiple choice 10 questions x 1mark) 10 Points
- 2) Subjective answer questions (6 questions x 5 marks) 30 Points

Total points	40
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10.3 External Practical Exam/Viva (20 Points):

Examination Division, Dean Office will conduct final practical examination at the end of semester.

10.4 Practical Exam/Viva (20 Points)

Internal assessment (Record Book-4 points, Project work Presentation- 2, Internal Practical Test- 2 Points)	Semester final examination	Total
8 Points	12 Points	20 Points

6. Recommended Books and References materials (including relevant published articles in national and international journals)**Prescribed Texts**

1. Charles S. Wasson, *System Engineering Analysis, Design, and Development*, Wiley

References

1. Jefferey A. Hoffer, Joey F. George, Joseph S. Valacich, *Modern Systems Analysis and Design*, 7th Edition, Pearson Education
2. Jeffrey L. Whitten, Lonnie Bentley, *System Analysis and Design methods*, 7th Edition, McGraw Hill
3. Joseph S. Valacich, Joey F. George, Jefferey A. Hoffer, *Essentials of System Analysis and Design*, 5th Edition, Pearson Education.

4. Gary B. Shelly, Harry J. Rosenblatt, System Analysis and Design, 9th Edition, Shelly Cashman Series
5. Alan Dennis, Barbara Haley Wixom, Roberta M. Roth System Analysis and Design, 4th Edition, Wiley Publication
6. V. Rajaraman, Analysis and Design of Information System, 2nd Edition, Prentice Hall